



Corrected TD2: Stabilizers

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1 Introduction

$$J(\theta) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & e^{i\theta} \\ 1 & -e^{i\theta} \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

Clifford operation U	Pauli input P	Pauli output $UPU^\dagger$
$CNOT_{12}$	X_1	X_1X_2
	X_2	X_2
	Z_1	Z_1
	Z_2	Z_1Z_2
H	X	Z
	Z	X
S	X	Y
	Z	Z

Exercise 1 (Introduction to stabilizers).

1. What are the result of the following products:

- $SXS^\dagger = Y$
- $SZS^\dagger = Z$
- $Cx_{12}I_1 \otimes X_2Cx_{12} = X_2$
- $Cx_{12}X_1 \otimes I_2Cx_{12} = X_1X_2$
- $Cx_{12}I_1 \otimes Z_2Cx_{12} = Z_1Z_2$
- $Cx_{12}Z_1 \otimes I_2Cx_{12} = Z_1$
- $HYH = -Y$
- $SYS^\dagger = -X$

2. Show that $J(0)J(\alpha) = P(\alpha)$ and using $J(\pi)$ and $J(0)$ show that $X = J(\pi)H$

$$P(\alpha) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & e^{i\alpha} \end{pmatrix}$$

Exercise 2 (Manipulating stabiliser states).

1. What state is stabilised by the stabiliser group $S = \langle Z_1, Z_2, Z_3 \rangle$. We call this state $|\Psi_0\rangle$
 $|000\rangle$
2. What is the stabiliser group of the following states:

- $S_3 C_{z_{12}} C_{z_{23}} H_2 H_1 |\Psi_0\rangle < XZI, ZXZ, ZXI >$
- $C_{x_{13}} C_{x_{12}} H_1 |\Psi_0\rangle < XXX, ZZI, ZIZ >$

3. Write the following state in the canonical basis ($|000\rangle, |001\rangle, \dots$) and find its stabiliser group: $C_{z_{12}} C_{z_{23}} H_3 H_2 H_1 |\Psi_0\rangle$
 $|\Psi'\rangle = |000\rangle + |001\rangle + |010\rangle + |100\rangle - |011\rangle + |101\rangle - |110\rangle + |111\rangle$ and $S = < XZI, ZXZ, IZX >$
4. If you have a n-qubit state, how many terms is there at worst in the vector state representation? How many stabiliser generator is there at worst in the stabiliser formalism ?
 Either you can represent your state with 2^n terms, or you can represent it with n independent Pauli operators from P_n